



A further experimental and modeling study of acetaldehyde combustion kinetics

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ABSTRACT

Acetaldehyde is an important intermediate and a toxic emission in the combustion of fuels, especially for biofuels. To better understand its combustion characteristics, a detailed chemical kinetic model describing the oxidation of acetaldehyde has been developed and comprehensively validated against various types of literature data including laminar flame speeds, oxidation and pyrolysis in shock tubes, chemical structure of premixed flames, and low-temperature oxidation in jet-stirred reactors. To extend the validation range, the chemical structure of a counterflow flame fueled by acetaldehyde at 600 Torr has been measured using vacuum ultra-violet photoionization molecular-beam mass spectrometry. In addition, ignition delay times at 10 atm and 700–1100 K were measured in a rapid compression machine, and a negative temperature coefficient (NTC) behavior was observed. The present kinetic model well reproduces the results of various acetaldehyde combustion experiments covering wide ranges of temperatures (300–2300 K) and pressures (0.02–10 atm), and explains well the observed NTC behavior based on the competition between multiple oxidation pathways for the methyl radicals and their self-recombination forming ethane, a relatively stable species at temperatures below 1000 K.

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1. Introduction

Acetaldehyde plays an important role in biofuel combustion as one of the most abundant toxic, oxygenated emissions [1,2]. For example, experimental studies on the structure of the premixed flames fueled by butanols show that acetaldehyde concentration could be as high as 6000 ppm [3]. A comprehensive chemical kinetic mechanism describing the oxidation and pyrolysis behavior of acetaldehyde is crucial to the construction of combustion kinetic models for practical fuels, especially for biofuels, and can also contribute to improved understanding and prediction of pollutant formation.

Several kinetic models have been proposed to describe the oxidation or pyrolysis of acetaldehyde. Specifically, the model of Kaiser et al. [4] described acetaldehyde oxidation in the negative temperature coefficient (NTC) regime between 550 and 900 K, while the model of Pelucchi et al. [5,6] well reproduced the

observed ignition and cool flame behaviors of mixtures of acetaldehyde and oxygen from 450 to 625 K [7]. Dagaut and co-workers [8] proposed a model to describe the oxidation in a jet-stirred reactor (JSR) and the ignition in a shock tube (ST) covering medium to high temperature ranges (900–2530 K). A high-temperature model developed by Leplat and Vandooren [9] was validated against their speciation measurements in low-pressure premixed flames. The AramcoMech 2.0 [10–14], developed as a core mechanism, has been tested against the ignition delay times of acetaldehyde measured by Yasunaga et al. [15] and the speciation results during the oxidation at 1100 K in a flow tube [16]. Based on theoretical calculations, Sivaramakrishnan et al. [17] built a model for the pyrolysis of acetaldehyde. In spite of such extensive and worthy efforts, it is nevertheless noted that they were all developed based on only one or two sets of experimental results, hence limiting the validation range for parameters. Mével et al. [18] published a widely-validated kinetic model, tested against the oxidation and pyrolysis results in STs [15,19,20], laminar flame speeds [21], oxidation in JSR [8] and speciation measurement in low-pressure premixed flames [22]. This model, however, focuses on the high-temperature pyrolysis and oxidation

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of acetaldehyde, containing no reactions related to the low-temperature oxidation. Recently, speciation of the acetaldehyde oxidation in a JSR was measured by Zhang et al. [23] between 460 and 900 K. A model for acetaldehyde oxidation was also proposed by incorporating theoretically calculated rate coefficients for the H-abstractions from CH_3CHO by CH_3O , CH_3O_2 and CH_3CO_3 radicals. In their model, a pathway of second O_2 -addition to CH_3CO radical was included to explain the high reactivity of acetaldehyde oxidation at low temperatures, although, this pathway was theoretically found by Maranzana et al. [24] to be negligible in the oxidation of CH_3CO radicals. A comprehensively validated kinetic model for acetaldehyde combustion that covers the combined low- and high-temperature chemistries is therefore needed.

In terms of experimental investigations, it is noted that over the past decades, the oxidation and pyrolysis of acetaldehyde have been experimentally investigated in STs [8,15,18–20,25,26], laminar premixed flames [9,21,22], JSRs, and flow reactors [16,27]. In 1981, Gray et al. [7] reported the oscillatory ignition and cool flames phenomenon in the acetaldehyde oxidation. Subsequently, Kaiser et al. [4] investigated the NTC behavior in the low-temperature oxidation of acetaldehyde, providing information about the oxidation process based on their measurements at two temperatures (553 and 713 K). Recently, Tao et al. [28] and Zhang et al. [23] separately reported the species concentration profiles measured in the low-temperature oxidation of acetaldehyde in JSRs, thus broadening the kinetic understanding of the NTC behavior of acetaldehyde. The NTC behavior of various fuels has been found [29–32] in the measured ignition delay times, but until now, such observation has not yet been reported for acetaldehyde. It is therefore meaningful to investigate the manifestation of the NTC phenomenon in the ignition delay times and explore the associated oxidation kinetics.

As for the high-temperature oxidation of acetaldehyde, detailed experimental information is also limited. Besides the measurements of the global characteristics, such as the laminar flame speeds [21] and the ignition delay times [8,15,18,25], Leplat and Vandooren [9] measured the mole fraction profiles of five intermediates in the laminar premixed flames of acetaldehyde. We recently extended this flame work by employing vacuum ultra-violet photoionization molecular-beam mass spectrometry (VUV-PI-MBMS) to determine the detailed chemical structure of two low-pressure laminar premixed flames [22], recognizing nevertheless that these low-pressure flame results are not likely to be sufficient to unambiguously examine the high-temperature sub-mechanism of acetaldehyde combustion. Detailed experimental results from different flame structures or under different pressures, for example as those in the counterflow flames, would facilitate the validation of a comprehensive model.

In the present study, a kinetic model for the oxidation of acetaldehyde was developed as a continuing effort of our previous work [28], in which a modified model based on AramcoMech 2.0 was proposed to describe and explain the low-temperature oxidation of acetaldehyde, especially the NTC behavior. The performance of the present kinetic model has been comprehensively examined with several sets of experimental data in the literature [8,15,21,22,26,28]. Furthermore, to extend the validation range, the ignition delay times at medium temperatures (700–1100 K) were measured in a rapid compression machine (RCM) and the species concentration profiles in a counterflow flame fueled by acetaldehyde were measured by VUV-PI-MBMS.

2. Experimental specifications

2.1. Rapid compression machine

The ignition delay times of acetaldehyde were measured in the RCM at Tsinghua University, covering the temperature range of

Table 1

Mixture compositions and experimental conditions.

	Equivalence ratio	Mole fractions (%)				Pressure bar	Temperature K
		CH_3CHO	O_2	Ar	N_2		
1	1.0	2.78	6.92	90.30	/	10	734–1086
2	–	2.78	/	90.25	6.97		

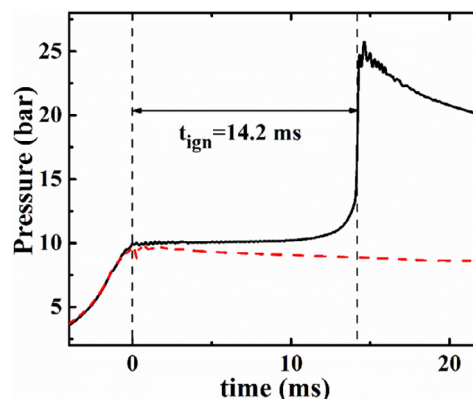


Fig. 1. Pressure trace history of reactive (black solid line) and nonreactive mixture (red dashed line) in the RCM at $P_{\text{eff}} = 10$ bar, $T_{\text{eff}} = 851$ K, $\Phi = 1.0$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

734–1086 K at 10 bar with the equivalence ratio of 1.0. The configuration of the RCM has been described in details by Di et al. [33]. The gaseous mixture was prepared in a stainless steel mixing tank, whose composition was determined according to the partial pressure of each mixture component. The partial pressure of acetaldehyde was controlled to be less than 50% of its vapor pressure to avoid potential condensation. Argon was selected as the dilution and the diluent ratio (Ar/O_2) was fixed at 13 to obtain a wide test temperature range. The mixture composition and experiment conditions are listed in Table 1. To calibrate the heat-loss effect after the compression, a nonreactive mixture with similar heat capacity was prepared by replacing O_2 with N_2 , as shown in Table 1. Since the partial pressure of each component cannot be controlled exactly in the preparation of each mixture, there is a slight difference in the concentrations of the two mixtures. Pressure traces were recorded for both mixtures under the same conditions.

Figure 1 displays a typical pressure trace profile using both mixtures at the same condition. Different from the situations for other fuels, the heat release rate of acetaldehyde oxidation before ignition is larger than the heat loss rate caused by heat conduction, due to the high activity of acetaldehyde oxidation especially at temperatures lower than 900 K. Consequently, after the end of compression (EOC), the pressure increases slowly, rather than decreasing as in the cases for common fuels. The instant of EOC was determined as the first peak pressure after compression in the recorded pressure history of the nonreactive mixture. The effective pressure ($P_{\text{eff}} = 10$ bar) was determined as the pressure at the EOC point instead of the integrated average of pressure from the first peak pressure to the minimum pressure after compression, as explained in Refs [33,34]. Based on the isentropic assumption, the effective temperatures ($T_{\text{eff}} = 734$ –1086 K) were calculated from the pressure by considering the corresponding compression ratios. The ignition delay time was defined as the time interval between EOC and the state of the maximum rate of the pressure rise. The method for the uncertainty evaluation has been described in Ref. [33]. The measured ignition delay times with their

Table 2

List of reactions and the corresponding kinetics parameters implemented in the current kinetic model, $k = A T^n \exp(-E_a/RT)$, the unit for T is K.

	Reaction	A $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$	n	E _a cal/mol
R1	$\text{CH}_3\text{CHO} + \text{HO}_2 = \text{CH}_3\text{CO} + \text{H}_2\text{O}_2$	5.560×10^{-3}	4.5	4829
R2	$\text{CH}_3\text{CHO} + \text{HO}_2 = \text{CH}_2\text{CHO} + \text{H}_2\text{O}_2$	1.086×10^{-1}	4.1	18,648
R3	$\text{CH}_3\text{CHO} + \text{CH}_3 = \text{CH}_3\text{CO} + \text{CH}_4$	1.550×10^0	3.7	4564
R4	$\text{CH}_3\text{CHO} + \text{CH}_3 = \text{CH}_2\text{CHO} + \text{CH}_4$	1.437×10^0	3.7	8857
R5	$\text{CH}_3\text{CHO} + \text{CH}_3\text{O}_2 = \text{CH}_3\text{CO} + \text{CH}_3\text{O}_2\text{H}$	3.218×10^{-1}	3.94	9503
	DUP	4.990×10^{-6}	4.98	5268
R6	$\text{CH}_3\text{CHO} + \text{CH}_3\text{O}_2 = \text{CH}_2\text{CHO} + \text{CH}_3\text{O}_2\text{H}$	8.230×10^{-3}	4.28	16,817
	DUP	7.080×10^{-9}	5.66	11,699
R7	$\text{CH}_3\text{CHO} + \text{CH}_3\text{CO}_3 = \text{CH}_3\text{CO} + \text{CH}_3\text{CO}_3\text{H}$	1.916×10^0	3.64	5642
	DUP	9.810×10^{-4}	4.32	2636
R8	$\text{CH}_3\text{CHO} + \text{CH}_3\text{CO}_3 = \text{CH}_2\text{CHO} + \text{CH}_3\text{CO}_3\text{H}$	2.540×10^{-3}	4.36	13,426
	DUP	3.600×10^{-11}	6.20	7351
R9	$\text{CH}_3 + \text{CH}_3(+M) = \text{C}_2\text{H}_6(+M)$	$9.500 \times 10^{+14}$	-0.538	179
	Low	$1.269 \times 10^{+41}$	-7.0	2762
	Troe 0.620	73	1180	$1.0 \times 10^{+30}$
R10	$\text{CH}_3 + \text{H}(+M) = \text{CH}_4(+M)$	$2.300 \times 10^{+04}$	0.032	144
	Low	$2.700 \times 10^{+35}$	-5.345	3380
	Troe 0.395	163.5	4250.3	1,253,976
R11	$\text{CH}_3 + \text{HO}_2 = \text{CH}_4 + \text{O}_2$	$2.320 \times 10^{+05}$	2.23	-3022

corresponding uncertainties are provided in the Supplemental Material.

2.2. Counterflow flame

The counterflow flame experiments were performed at the Chemical Dynamics Beamline of the Advanced Light Source (ALS) of the Lawrence Berkeley National Laboratory. Only a brief description is given here as the details of the experimental setup are described elsewhere [35–37]. The experimental system consisted of two opposing burners, placed at a distance of 12 mm from each other. Each of them consisted of two concentric tubes with inner diameters of 12.5 and 20 mm, respectively. The argon-diluted fuel or oxygen flowed through the inner tube while the outer rings supplied pure argon to shield and stabilize the flame. Gas mixture was sampled from the flame along the centerline of the burners by a quartz nozzle placed horizontally to the centerline, then guided into a reflectron time-of-flight mass spectrometer (TOF-MS) employing the tunable synchrotron vacuum ultra-violet (SVUV) radiation as the source for photoionization (PI).

The oxygen/argon and fuel/argon mixtures were supplied from the upper and lower burners separately, with the flow rates of O_2 (0.20 slm)/ Ar (1.10 slm) and CH_3CHO (0.25 slm)/ Ar (1.00 slm), respectively. The counterflow flame was stabilized at 600 Torr. The mass spectra, which can be subsequently converted to the concentrations of different species, were collected along the centerline as functions of distance from the fuel outlet. For the species identification, the photoionization efficiency (PIE) curves were measured from 9.0 to 10.7 eV at 5.2 mm far from the surface of the fuel-side burner, where the concentrations of most intermediates reached maximum [38]. Temperature profiles were measured based on its relation with the pressure in the first stage chamber, which has been utilized as an accurate method for correcting the probe disturbance in the flame sampling experiments [37,39]. The concentration and temperature profiles are provided in the Supplemental Material, and the detailed information such as the photoionization cross sections used for quantification and uncertainties of the mole fraction profiles are presented in Table S1. The uncertainties of the mole fraction profiles were evaluated through the same method used in premixed flame studies [22,40,41], considering numerous effects, such as precision of the photon energy, quenching of the radicals caused by interaction with nozzle, mass discrimination, and uncertainty of the PICS data.

3. Modeling description

In our previous work [28], a kinetic model was developed based on AramcoMech 2.0 [10–14] to explain the low-temperature reactivity and the observed NTC behavior of acetaldehyde oxidation. In that model, the consumption pathways for the CH_3CO_3 radical, which is the key intermediate for the low-temperature oxidation of acetaldehyde, were supplemented and modified based on the theoretical work of Lee et al. [42]. In addition, the sub-mechanisms for ketene (CH_2CO), formic acid (HOCHO) and acetic acid (CH_3COOH) were also updated based on previous works [43–46].

In a continuation of this effort, the high-temperature and pyrolysis kinetics of the previous model are now updated partly by adopting the CH_3CHO pyrolysis mechanism proposed by Sivaramakrishnan et al. [17], which was built to reproduce the concentration histories of the pyrolysis products of CH_3CHO in shock tubes. Based on their work, additional modifications are also made in the present model, including the consumption pathways for the CH_2CHO radical, the isomerization and consumption of ethenol ($\text{C}_2\text{H}_3\text{OH}$), and some reactions between small hydrocarbons. Table 2 lists some key reactions which will be discussed in the following.

Hydrogen abstractions from CH_3CHO by different radicals are important chain-propagation reactions in acetaldehyde combustion. The rate parameters for the H-abstractions by H radicals are kept the same as in AramcoMech 2.0, i.e., taken from recent theoretical work of Harding et al. [47]. In our previous model [28], the H-abstractions by OH radicals were updated to improve the modeling predictions based on the theoretical work of Taylor et al. [48]. In the pyrolysis model of Sivaramakrishnan et al. [17], the two types of H-abstractions by the HO_2 radicals (R1 and R2 as listed in Table 2) were not included because there is no need to consider the HO_2 radical in a pyrolysis process. While in AramcoMech 2.0, only the H-abstractions from the aldehyde group by HO_2 radicals (R1) were considered. In addition, for the H-abstractions by the CH_3 radicals (R3 and R4), only the H-abstraction from the aldehyde group (R3) was considered in the pyrolysis model of Sivaramakrishnan et al. [17] and in AramcoMech 2.0. Consequently, the two missing H-abstractions from the methyl group by the HO_2 radical and the CH_3 radical (R2 and R4) in AramcoMech 2.0 are added in the present model based on the theoretical calculations of Mendes et al. [49], and the two corresponding reactions (R1 and R3) are also updated us-

Table 3

Conditions of the experimental data used for the model validation.

Experimental target	Reactor	Temperature K	Pressure atm	Equivalence ratio	Ref.
<i>Experimental data available in the literature</i>					
Flame speciation	Premixed flame	500–2300	0.02, 0.04	1.0, 1.7	[22]
Flame speed	Heat flux	$T_0 = 298$ –358	1.0	0.6–1.8	[21]
Ignition delay time	ST	1250–1700	2.03–2.32	0.5–2.5	[15]
Pyrolysis	ST	1000–1600	1.26–2.82	∞	[15,26]
Oxidation	JSR	500–950	0.92	0.5	[28]
Oxidation	JSR	950–1350	1, 10	0.09–1.61	[8]
<i>Experimental data obtained in this study</i>					
Flame speciation	Counterflow flame	330–1550	0.79	–	This work
Ignition delay time	RCM	700–1100	10	1.0	

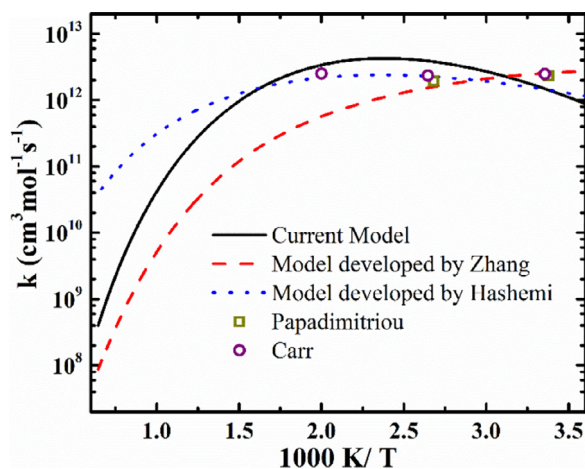


Fig. 2. Plot of the rate constants at 1 atm for $\text{CH}_3\text{CO} + \text{O}_2 = \text{CH}_3\text{CO}_3$. The black solid line represents the rate constant used in the present kinetic model provided by Lee et al. [42]. The red dashed line and blue dotted line represent the rate constant used in the model developed by Zhang et al. [23] and Hashemi et al. [56]. The gray square dots and purple circle dots are experimental results measured by Papadimitriou et al. [51] and Carr et al. [54]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ing the same source. The H-abstractions by CH_3O_2 and CH_3CO_3 radicals (R5–R8), especially for the H-abstractions from the aldehyde group (R5 and R7), were found to be important in the low-temperature oxidation of acetaldehyde in our previous work [28]. In the present model, their rate parameters are updated based on the recent theoretical results of Zhang et al. [23].

Since, the O_2 -addition to the fuel radical is crucial for chain branching in the low-temperature oxidation [50], we briefly review herein the O_2 -addition to the dominant fuel radical CH_3CO (R12). The $\text{CH}_3\text{CO} + \text{O}_2$ reaction system has been widely investigated in the area of atmospheric chemistry because of its importance in the consumption of several important volatile organic pollution [42,51–55], although most of them only provided the rate constants around room temperature. Specifically, Lee et al. [42] published a set of rate parameters for R12 based on their theoretical calculations, which was adopted in our previous kinetic model [42]. Hashemi et al. [56] employed another set of rate parameters in their model developed for methane oxidation, by fitting the theoretical results from Allen et al. [53]. Zhang et al. [23] used a different set of values in their model for acetaldehyde combustion based on the theoretical work of Carr et al. [54].



The different rate constants at 1 atm are compared with the literature values [51,54], as shown in Fig. 2. It is seen that the values used in different models are close to the measured values around room temperature, but have large discrepancy as temperature in-

creases. Within the specific temperature range between 700 and 1000 K, the rate constants from the three different models can differ up to two orders of magnitude, but no literature data are available for validation within this temperature range. Consequently, the rate constants provided by Lee et al. [42], which fall in the mid-range of the different calculated values, are adopted in the present model.

Three reactions related to the methyl radical (R9, R10, and R11) are also modified in the present model, because they are found to be important in the simulations of both RCM and counterflow flame experiments. The rate parameters of self-recombination of methyl radicals (R9) are updated using the values recommended by Baulch et al. [57], and the rate parameters of R10 are taken from the theoretical investigation of Troe and Ushakov [58]. The reaction parameters for R11 are modified based on the experimental data in the literature [59].

This latest kinetic model, consisting of 498 species and 2762 reactions, is provided in the Supplemental material along with the thermodynamic and transport data. A comprehensive validation has been conducted using the experimental data obtained in the present work as well as those from the literature, as summarized in Table 3 and discussed in details in the following section.

The simulations were performed with the Chemkin Pro Software [60] employing the corresponding modules for different types of experiments. Simulations for the oxidation and pyrolysis in the ST and RCM experiments were performed using the closed zero-dimensional homogeneous reactor code assuming constant volume and adiabatic condition. The counterflow and premixed flame experiments were simulated with the OPPDIF and PREMIX modules, respectively, taking the respective flame conditions and the measured temperature profiles as inputs. Simulation of the flame speed was also performed using the PREMIX module at different unburnt gas temperatures separately. Multi-component transport and thermal diffusion were considered in the simulations for all the flames. The transient perfectly stirred reactor code was used to simulate the low-temperature oxidation in the JSR with a fixed volume of 33.5 cm³ taking the recorded temperatures and corresponding inlet flow rates as input files.

4. Results and discussion

The following discussion can be divided into two parts. First, predictions of the present kinetic model are compared with the new RCM and counterflow flame experimental results. Subsequently, predictions of the current model are validated against the literature data to demonstrate its capability. Rate of production (ROP) analysis and sensitivity analysis are also performed and presented.

4.1. Low-temperature oxidation in RCM

Figure 3(a) compares the measured ignition delay times with the modeling predictions by AramcoMech 2.0 and the present

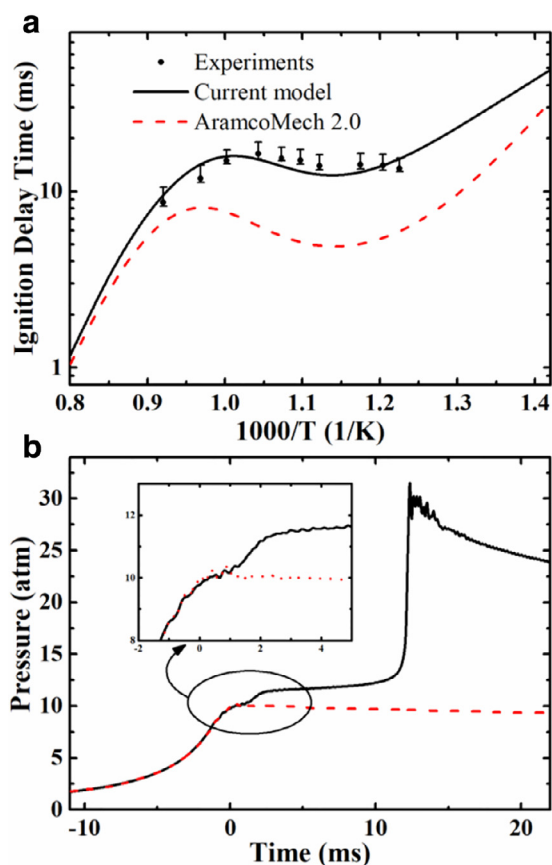


Fig. 3. (a) Comparisons of acetaldehyde ignition delay times. Symbols: experimental data. Lines: AramcoMech 2.0, and current model. (b) Pressure traces of experiments with (black solid line) without (red dashed line) O_2 at 735 K. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

model. It is seen that predictions of the present model are consistent with the measured values. Interestingly, a NTC behavior is observed in the experiments, which is captured by both models. In addition, a two-stage ignition, as the example of the recorded pressure traces at 735 K shown in Fig. 1(b), is clearly observed at three tested temperatures (735 K, 746 K, 757 K). The observation of the two-stage ignition proves the relatively high oxidation reactivity of

acetaldehyde at low temperatures and indicates the existence of its low-temperature oxidation pathways. However, the first-stage ignition occurs too close to the termination of compression, making it hard to exactly locate the termination of compression and determine the ignition delay times. Therefore, the ignition delay times at temperatures lower than 800 K are not shown in Fig. 3(a), and the corresponding pressure traces are provided in the Supplemental Material.

In our previous study on the low-temperature oxidation of CH_3CHO in JSR, the observed NTC behavior was explained by the competition between two different pathways for the acetyl radical (CH_3CO) consumption: the decomposition of CH_3CO and the O_2 -addition to CH_3CO [28]. However, it is a different story for the observed NTC behavior in the RCM experiments, because the NTC regime in the RCM experiments (800–1000 K) is at higher temperatures than that in the JSR experiments (600–750 K). To make the following discussion clear, the observed NTC behavior in the RCM and JSR experiments are referred as NTC-RCM and NTC-JSR, respectively. The formation of CH_3CO_3 , the key species in the explanation for NTC-JSR, is not favorable at temperatures higher than 800 K. Therefore, the explanation for NTC-JSR is not suitable for NTC-RCM.

To identify the reasons for the NTC-RCM, ROP analysis was performed at several temperatures for the conditions at which 20% of the fuel is consumed. Figure 4 summarizes the results at 700, 900, and 1150 K, which are respectively before, in the middle of and after the NTC-RCM temperature range. Similar to the NTC-JSR, the consumption of CH_3CHO mainly yields the CH_3CO radicals via H-abstractions by different radicals for all the investigated temperatures. For example, H-abstraction by the OH radical remains important at all tested temperatures between 700 and 1150 K, while H-abstraction by the CH_3CO_3 radical is important only at temperatures lower than 750 K.

Different from the case of the NTC-JSR, in the temperature range of the NTC-RCM, the O_2 -addition pathway is too weak to be noticed. Most of the CH_3CO radicals decompose into CH_3 radicals, which have three main consumption pathways: formation of CH_4 by taking one H atom from the HO_2 radicals, oxidation to CH_3O by HO_2 and self-recombination to C_2H_6 . Because all the formed CH_4 turn back to CH_3 radicals via H-abstractions by the OH and O radicals, the net production of CH_4 from the CH_3 radicals decreases with increasing temperature. As shown in Fig. 4, at all temperatures, a significant amount of the CH_3 radicals are oxidized by HO_2 to form CH_3O , which turn into CH_2O and finally yield CO and HO_2 . As the temperature increases, the importance of this CH_3O -related pathway decreases, while the production of C_2H_6

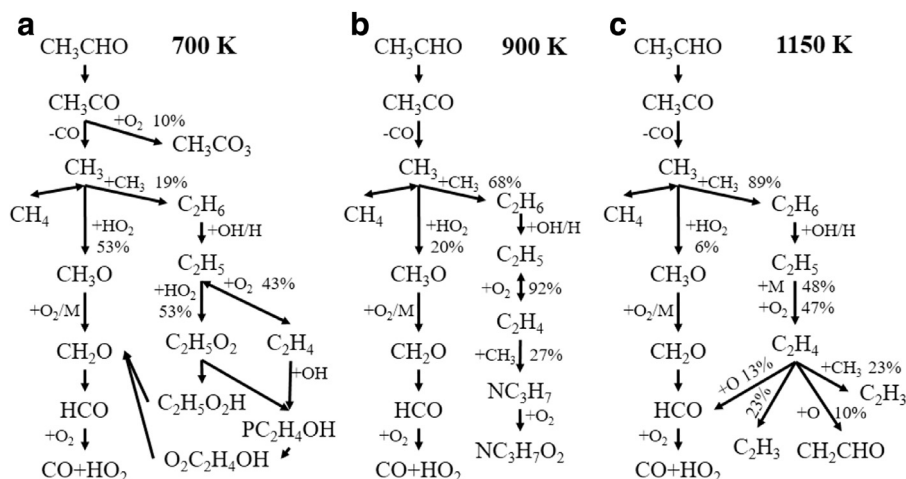


Fig. 4. ROP results using the current model for the consumption of CH_3CHO under (a) 700 K, (b) 900 K, and (c) 1150 K for the conditions at which 20% of the fuel is consumed.

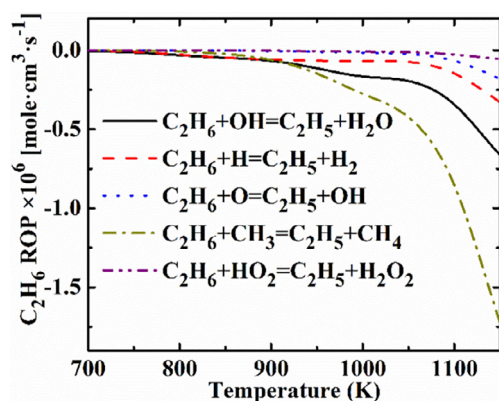


Fig. 5. Contributions of the top 5 reactions to the C_2H_6 consumption varying with temperature at 10 atm in the RCM experiment.

increases. At temperatures lower than 950 K, C_2H_6 is relatively stable, and is mainly consumed through some low-temperature oxidation pathways, as shown in Fig. 4(a) and (b). However, C_2H_6 becomes more reactive as temperature increases. As illustrated in Fig. 5, the consumption rate of C_2H_6 becomes faster as the temperature increases, especially for temperatures higher than 1000 K, which is the turning point of the NTC-RCM regime.

At 700 K, a small part of the formed CH_3CO radicals go through the O_2 -addition pathway forming the CH_3CO_3 radicals, which could explain the short first-stage ignition period observed at low temperatures in Fig. 3(b). This is consistent with our previous conclu-

sion that the CH_3CO_3 -related pathways result in the high reactivity of acetaldehyde oxidation at low temperatures. Most of the CH_3CO are consumed through the CH_3O -related pathway. As temperature increases, more CH_3 radical self-recombines into stable C_2H_6 . This change reduces the overall reactivity of CH_3CHO oxidation. With temperature increases further, C_2H_6 is not stable anymore. The rapid consumption of C_2H_6 restores the overall reactivity.

Therefore, the CH_3 radical is the key species to the ignition delay times of acetaldehyde, as pointed out by Dagaut et al. [8] based on their analysis of the shock tube experiments. Alkyl radicals not only rule the low-temperature oxidation of long chain aldehydes, as summarized by Pelucchi et al. [6], but also control the oxidation of CH_3CHO at medium temperatures. The different pathways for the CH_3 consumption at different temperatures have already been discussed to explain the NTC behavior in the predicted ignition delay time for CH_4 of Sabia et al. [61]. The NTC regime for CH_4 oxidation was observed [62] and predicted [61] to appear around 1000 K, which is similar to the temperature regime of the NTC-RCM for CH_3CHO oxidation, observed and predicted in the present work. Furthermore, the decrease in the reactivity for CH_4 oxidation in the NTC behavior is attributed to the recombination of CH_3 forming stable C_2H_6 in previous studies [61–63], as discussed in the last paragraph.

4.2. Mole fractions in the counterflow flame

Thirty-one species including 6 major species (Ar , O_2 , CH_3CHO , CO , CO_2 , and H_2O) were identified and quantified in the counterflow flame. As shown in Fig. 6(a), the current model predicts well

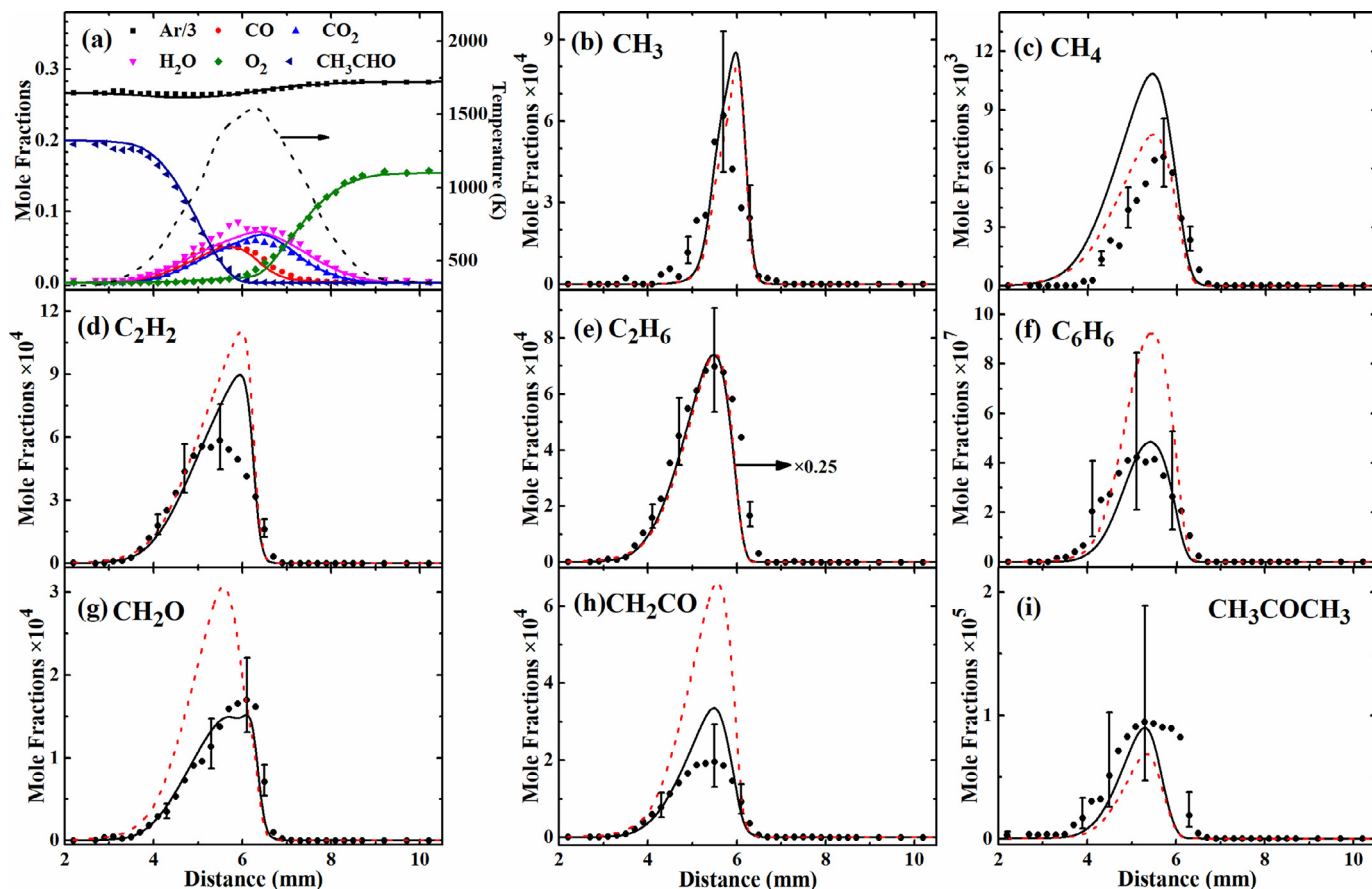


Fig. 6. Measured mole fraction profiles (symbols) of the major species and several intermediates in the counterflow flame compared with the predictions by current model (black solid lines) and AramcoMech 2.0 (red dashed lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

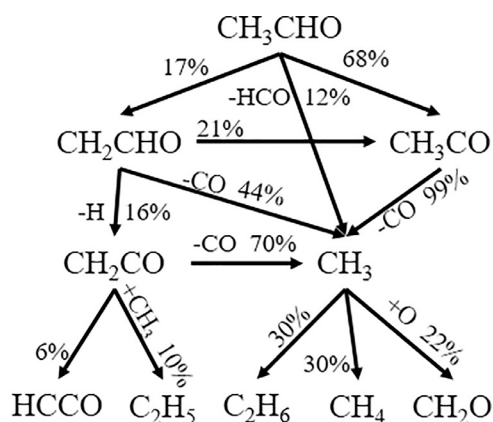


Fig. 7. Fuel consumption steps in counterflow flames predicted by the current model based on integrated ROP analysis along the entire flame. Numbers refer to the contributions of each reaction to consumption.

the measured mole fraction profiles of the major species. Compared with the predictions by AramcoMech 2.0, the agreement between model and experimental results for some important hydrocarbons and oxygenated intermediates are improved, as shown in Figs. 6 and S1. The detailed chemical information probed from the counterflow flame fueled by acetaldehyde demonstrates the satisfactory predictive capability of the present model.

H-abstractions from acetaldehyde, forming CH_3CO and CH_2CHO radicals, are the first step in the fuel consumption. As summarized in Fig. 7, subsequent to the H-abstractions, all the formed CH_3CO radicals decompose into CH_3 and CO . While for the CH_2CHO radicals, in addition to the direct decomposition pathway, they could

also go through isomerization turning into the CH_3CO radicals or be dehydrogenated forming ketene (CH_2CO), which mainly decomposes into CO and CH_3 by reacting with the H radicals. In addition, the direct decomposition of CH_3CHO giving CH_3 and HCO radicals is also important to the fuel consumption in the counterflow flame. Indeed, no matter which pathway the fuel molecules go through, they eventually turn into methyl radicals, which are well predicted by the present model, as can be seen in Fig. 6(b).

Figure 7 displays the three consumption pathways for the CH_3 radicals. As can be seen in Fig. 6(c), (g), the present model satisfactorily predicts the concentration profiles of methane and formaldehyde, which are two of the main products of CH_3 consumption. According to the ROP analysis based on the current model, the methane formed are consumed by going back to methyl radicals, while formaldehyde react with OH or H forming HCO and finally turn into CO . Self-recombination between the methyl radicals, contributing to all the formation of ethane, is an important reaction pathway in acetaldehyde counterflow flame, because it opens the reaction channel to the growth of larger hydrocarbons. As shown in Fig. 6(e), ethane is slightly over-predicted in the counterflow flame, however, we will see that the present model gives a satisfactory prediction for ethane in the following discussion about the premixed flames. The ethane chemistry in these two different flames are the same, i.e., entirely produced through self-recombination and consumed via H-abstractions by H , OH , O , and CH_3 radicals. Even the contributions from these four H-abstractions are similar in the two different flames. As shown in Fig. 6(d), (f) and Fig. S1, C_2H_2 , C_4 hydrocarbons, and even benzene are well predicted by the present model, but with the C_3 hydrocarbons being slightly over-predicted. The larger hydrocarbons are produced mainly via CH_3 -addition following the sequence from small hydrocarbons to larger hydrocarbons, starting from ethane step by step. The

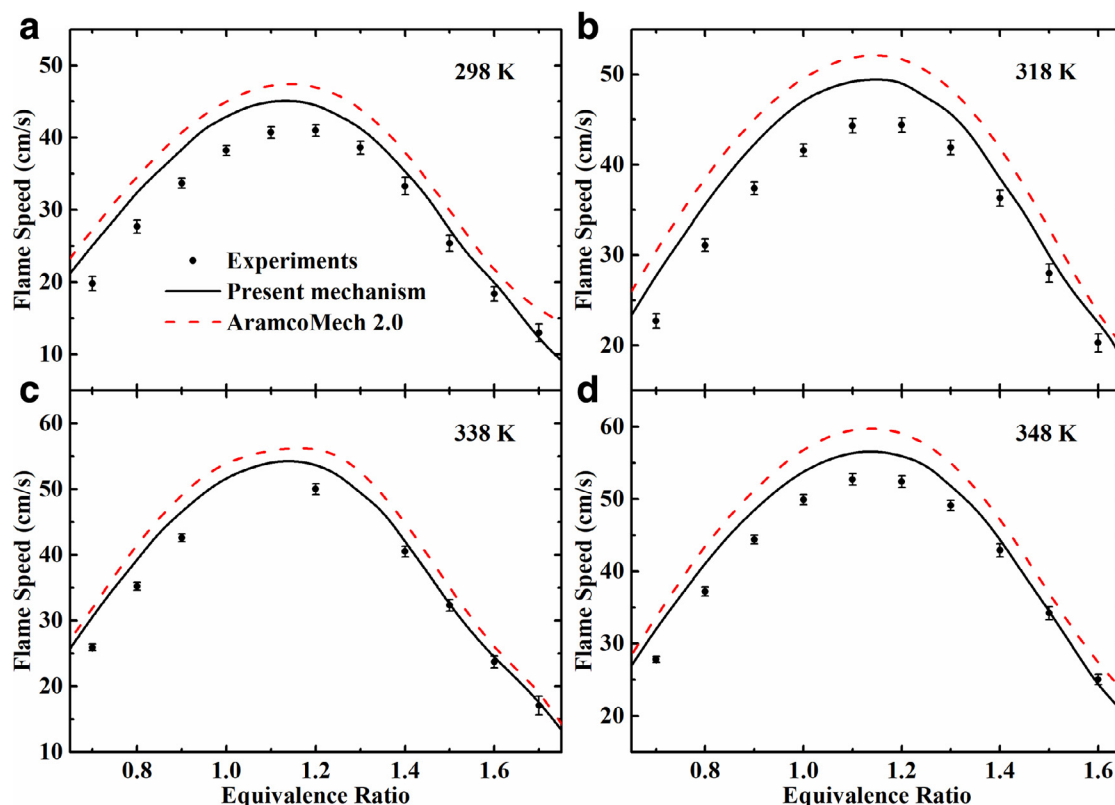


Fig. 8. Flame speeds of acetaldehyde/air mixture at 1.0 atm. Symbols: experimental results from Christensen et al. [21]; Lines: predictions using current model (black solid lines) and AramcoMech 2.0 (red dashed lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

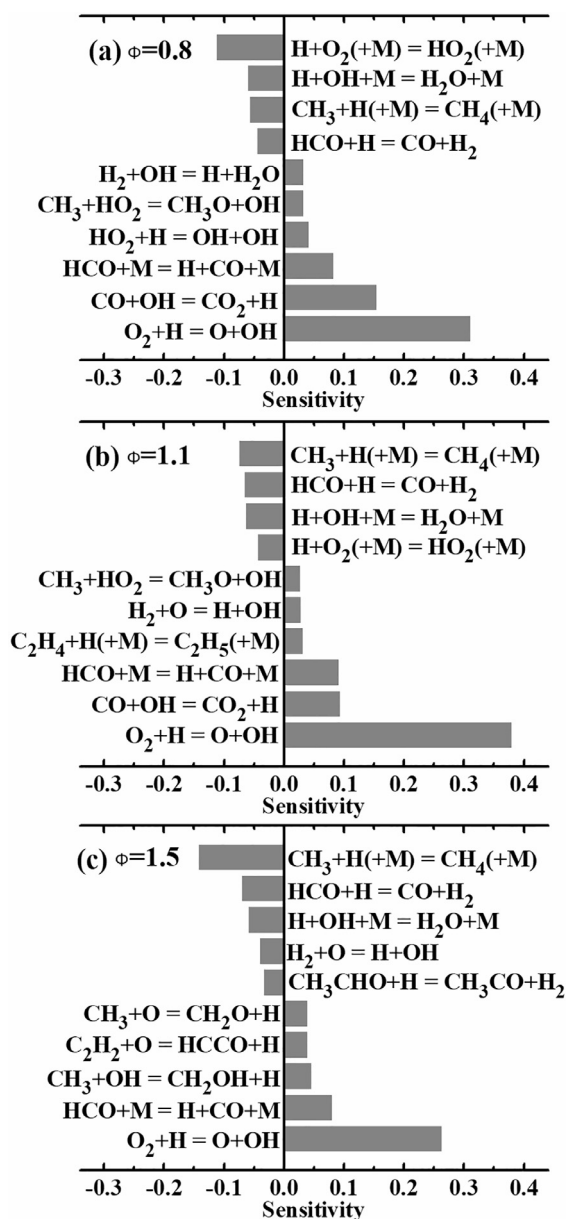


Fig. 9. Sensitivities of the flame speed of acetaldehyde and air with the unburned temperature of 318 K at different equivalence ratios using current model.

unsatisfactory prediction for ethane in the counterflow flame probably results from the part describing the reactions between the larger hydrocarbons in the present model, instead of the CH_3CHO related chemistry.

4.3. Validation against previous experimental data

As listed in Table 1, various kinds of experimental results from the literature are adopted in the present work for model validation, including laminar flame speed, oxidation and pyrolysis in ST, detailed chemical structure in the laminar premixed flames and the low-temperature oxidation in a JSR. In the following, we shall first present the validation against the global properties, such as flame speed and ignition delay times to show the overall performance of current model. Secondly, species mole fraction profiles measured in different reactors will be discussed to provide insights into the chemical details of acetaldehyde oxidation.

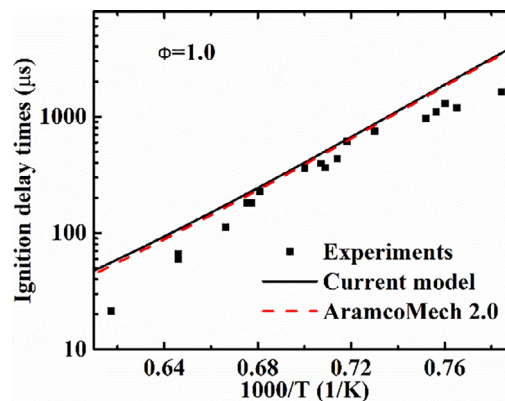


Fig. 10. Shock tube ignition delay times of acetaldehyde/oxygen/argon mixtures ($\Phi=1.0$, $P=2.06$ atm, 2% CH_3CHO / 5% O_2 / 93%Ar). Symbols: experimental results [15]; Lines: predictions by current model (black solid line) and AramcoMech 2.0 (red dashed line). For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

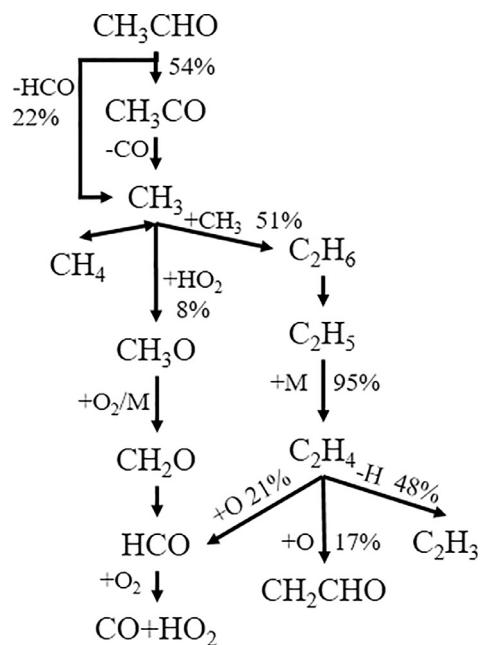


Fig. 11. ROP results using current model for the consumption of CH_3CHO at 1550 K for the conditions at which 20% of the fuel is consumed.

4.3.1. Laminar flame speed

Christensen et al. [21] measured the flame speeds of acetaldehyde and air mixture at different unburned temperatures with different equivalence ratios varying from 0.6 to 1.8. As shown in Fig. 8, compared to AramcoMech 2.0, predictions of the present model are improved at all investigated equivalence ratios and temperatures.

Sensitivity analysis were performed at three different equivalence ratios: fuel-lean ($\Phi=0.8$), fuel-rich ($\Phi=1.5$) and near-stoichiometric ($\Phi=1.1$) where the flame speed reaches its maximum. Results for the case with the initial temperature of 318 K are illustrated in Fig. 9. The reactions in the H_2/CO sub-mechanism, such as the reactions between H radical and O_2 , substantially affect the predicted flame speed. Among them, only reaction R13 and R14 are modified during the development of the present model based on the pyrolysis mechanism proposed by Sivaramakrishnan et al. [17]. In addition, the reaction between methyl and hydrogen radicals forming methane (R10) is also updated during the model development.

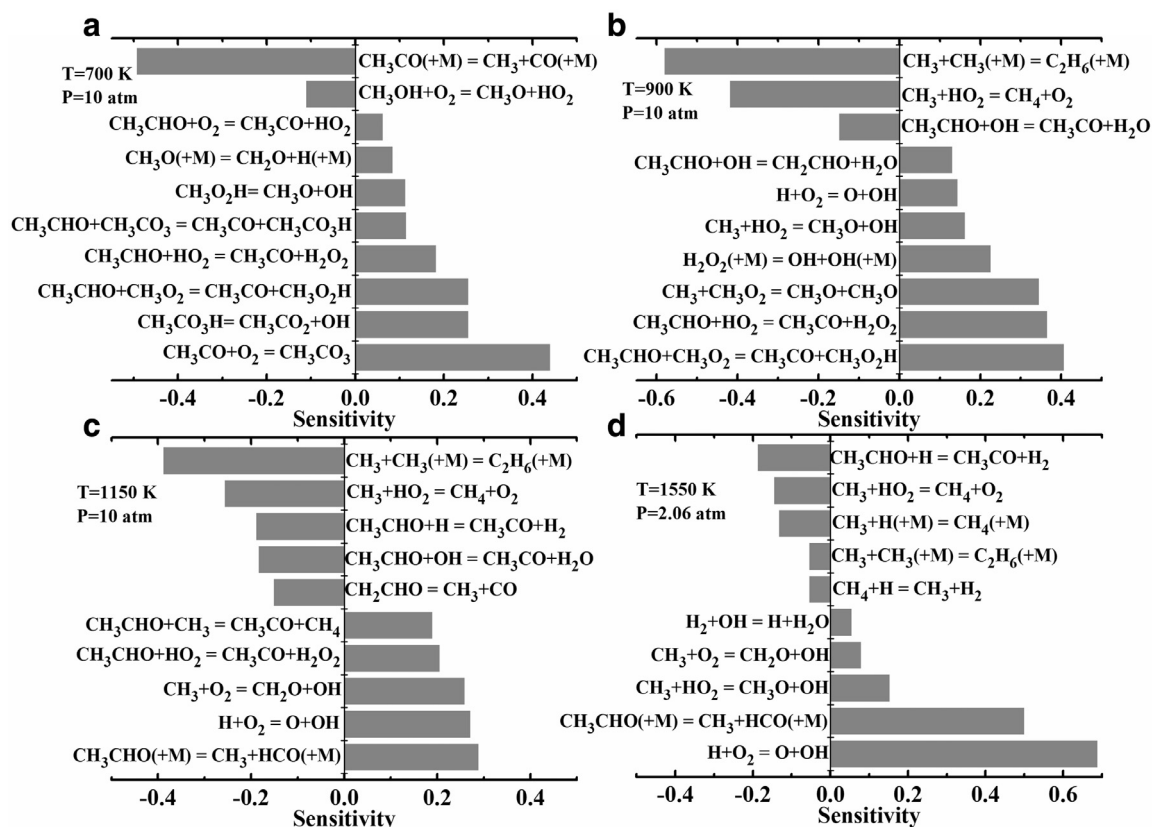


Fig. 12. Sensitivities of the ignition delay times of acetaldehyde at (a) 700 K, (b) 900 K, (c) 1150 K in RCM and (d) 1550 K in ST using the current model.

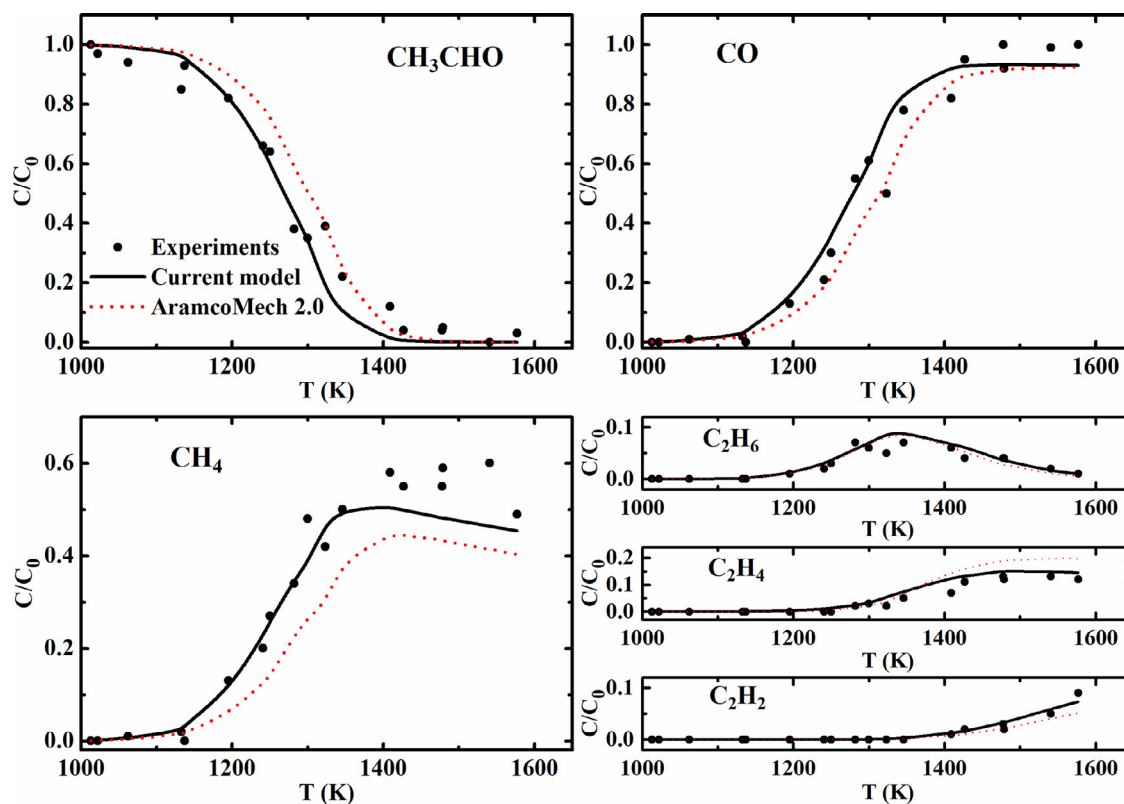


Fig. 13. Product distribution observed with mixture (5% CH₃CHO and 95% Ar) at the effective heating time at 50 Torr. Symbols: experimental results [15]; Lines: predictions by current model (black solid line) and AramcoMech 2.0 (red dot line). For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

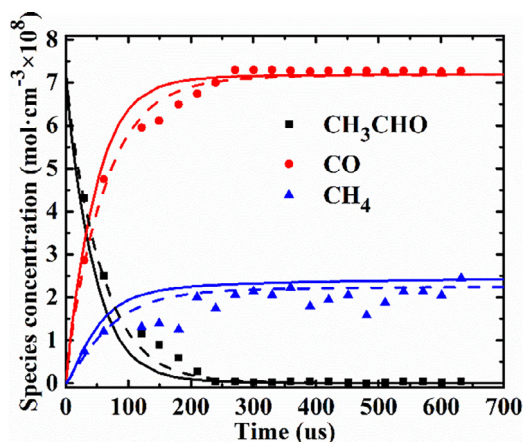


Fig. 14. Species profiles during the CH_3CHO pyrolysis in a shock tube. Symbols: experimental results [26]; Lines: predictions by current model (solid line) and AramcoMech 2.0 (dashed line).



Compared with the sensitivity results at the fuel-rich side ($\Phi = 1.1$ and 1.5), the importance of R14 is highlighted only in the sensitivity results at the fuel-lean side. This observation could explain the improvement of the predictions at the fuel-lean side than at the fuel-rich side of the current model.

4.3.2. Oxidation and pyrolysis in ST

The AramcoMech 2.0 has been validated against the ignition delay times at high temperatures measured by Yasunaga et al. [15]. The present model is also used to reproduce the experimental results to check its predictive capability. Only the results at equivalence ratio of 1.0 are presented here in Fig. 10; the other two sets at equivalence ratios of 0.5 and 2.5 are shown in Fig. S2. As it can be seen, the present model yields a satisfactory prediction for the ignition delay times at temperatures between 1250 and 1700 K.

The stoichiometric experimental results are chosen as an example here because this equivalence ratio has been adopted for the current RCM experiments. Due to the characteristics of RCM and ST, their suitable experimental conditions are different [29], therefore, we cannot measure the ignition delay times in the RCM under the same conditions as in Fig. 10. While, the pressures and concentrations in these two sets of experiments are somewhat different, these two sets of experimental results are still comparable and the comparison between them could identify the different be-

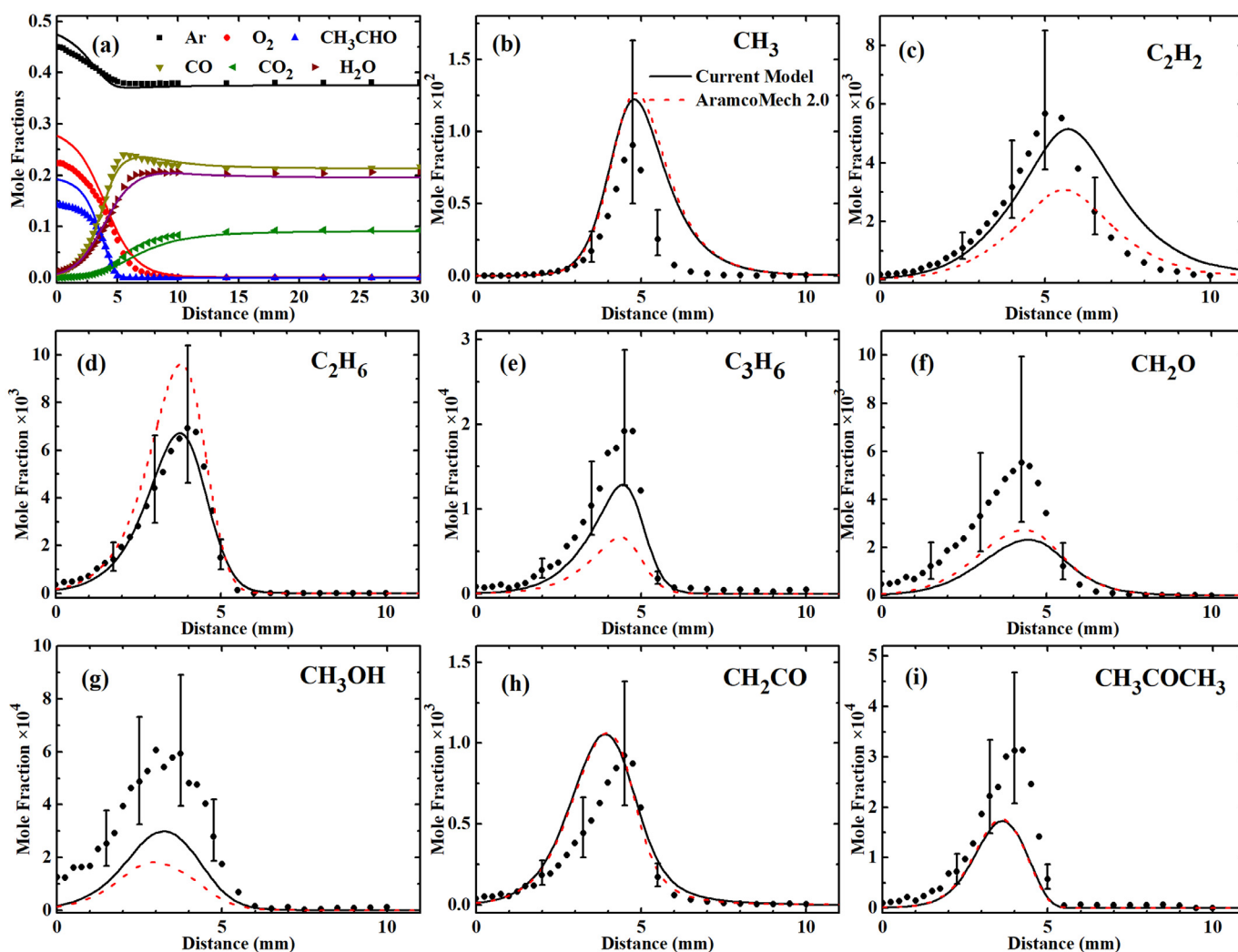


Fig. 15. Measured mole fraction profiles (symbols) of major species and several intermediates [22] compared with predictions by current model (black solid lines) and AramcoMech 2.0 (red dashed lines) in fuel-rich laminar premixed flame. For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

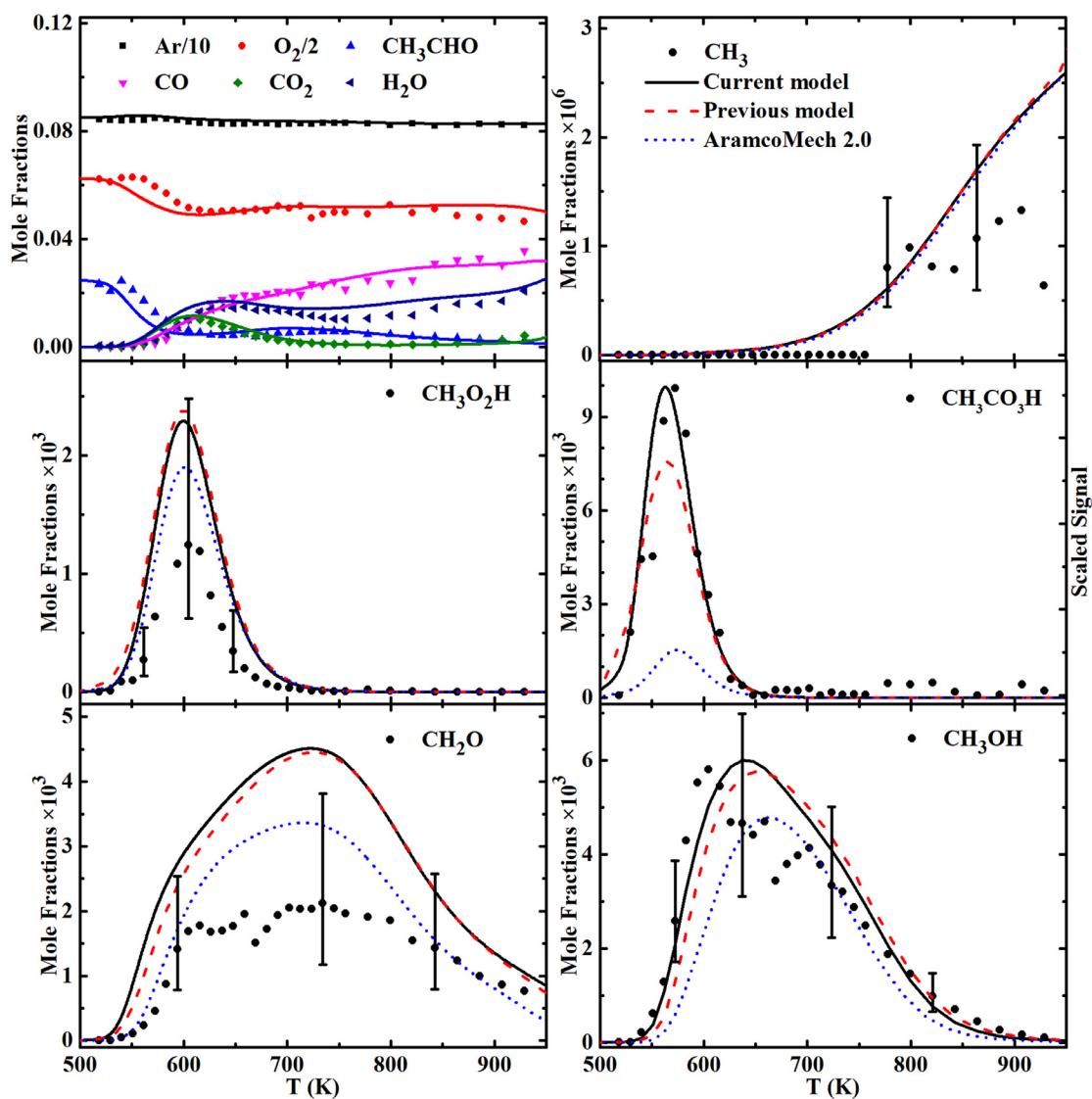
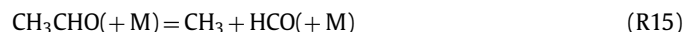


Fig. 16. Mole fraction profiles of major species, CH_3 , $\text{CH}_3\text{CO}_3\text{H}$, $\text{CH}_3\text{O}_2\text{H}$, CH_2O and CH_3OH in the low-temperature oxidation of acetaldehyde in JSR. Symbols: experimental results [28]; Lines: predictions by current model (black solid lines), previous model (red dashed lines) [28], and AramcoMech 2.0 (blue dot lines). For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

haviors of the model at different temperatures. Similar to Fig. 4, Fig. 11 shows the results of the ROP analysis for the oxidation in the ST at 1550 K ($1000/T=0.65$) for the condition at which 20% of the initial fuel concentration is consumed. Generally, the reaction pathways at 1550 K are similar as they are at 1150 or 900 K, despite the different contributions from different reactions. The most obvious difference is the appearance of the direct decomposition of acetaldehyde (R15), which makes a great contribution to the fuel consumption at 1550 K. While, R15 contributes only about 3% to the fuel consumption in the RCM at 1150 K, and hence it is not shown in Fig. 4(c).



Sensitivity analyses were performed at 700, 900, 1150 and 1550 K to determine the key reactions that have large influence on the predictions for the ignition delay times. At low temperatures (700 and 900 K), the ignition delay times are mainly sensitive to some reactions related to the peroxide intermediates, such as the H-abstractions by HO_2 , CH_3O_2 and CH_3CO_3 radicals, the O_2 -addition to CH_3CO radical and the decomposition of $\text{CH}_3\text{CO}_3\text{H}$. While at higher temperatures (1150 and 1550 K), the ignition delay

times are also sensitive to the reactions belonging to the CH_4 and H_2/O_2 sub-mechanism, in addition to the H-abstractions by H and OH radicals. At intermediate temperatures (900 and 1150 K), where the NTC behavior appears, the recombination of CH_3 radicals has significant effect on the predicted ignition delay times. As shown in Fig. 12(d), as an important consumption pathway at 1550 K, R15 is the second most sensitive reaction at 1550 K. Interestingly, although R15 does not significantly contribute to the fuel consumption at 1150 K, the ignition delay time at 1150 K is still sensitive to it, which confirms the validity of updating the pyrolysis sub-mechanism in the current model.

In addition to ignition delay times, Yasunaga et al. [15] also measured the concentration profiles of some products in the pyrolysis of CH_3CHO as functions of the temperature in a shock tube. As illustrated in Fig. 13, the present model could also well reproduce these experimental results.

Product concentration histories were also measured by Kern et al. [26] during the pyrolysis of a mixture of 3.2% CH_3CHO and 96.8% Ne in a shock tube at 0.32 atm. This set of experimental data was used by Sivaramakrishnan et al. [17] to validate their model, which is a major basis in the development of the current model.

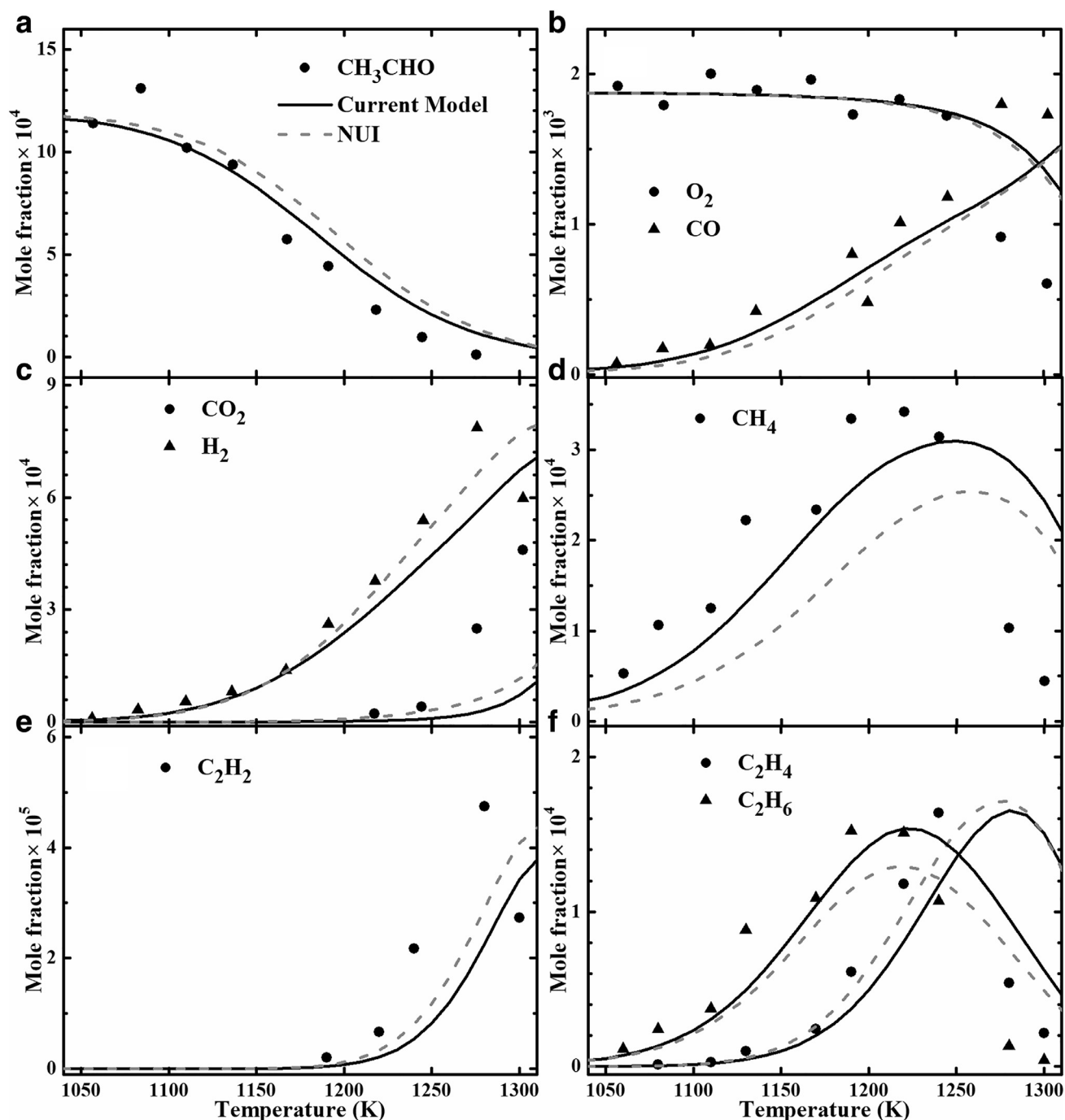


Fig. 17. Mole fraction profiles of major species, CH_4 , C_2H_2 , C_2H_4 , and C_2H_6 in the low-temperature oxidation of acetaldehyde in JSR at 1 atm, $\Phi = 1.0$, and residence time of 0.08 s. Symbols: experimental results [8]; Lines: predictions by current model (black solid lines) and AramcoMech 2.0 (Gray dash lines).

So, it is not surprising that the present model also well reproduces the experimental results, as shown in Fig. 14.

These two sets of detailed experimental data for the pyrolysis process therefore clearly demonstrate the satisfactory performance of the present model, at least the pyrolysis sub-mechanism.

4.3.3. Laminar premixed flame

The present model is further validated against the mole fraction profiles in the fuel-rich and stoichiometric laminar premixed flames measured by Tao et al. [22]. Since the comparisons between experiments and predictions for these two flames are similar, only the comparison of the major species and some intermediates in the fuel-rich flame are presented in Fig. 15.

In the case of the counterflow flame discussed above, both the H-abstractions and direct decomposition make significant contributions to fuel consumption. Similarly, in the premixed flames, CH_3CHO is mainly consumed via H-abstractions. However, due to the different structures of the premixed and non-premixed flames, the direct decomposition in the premixed flames is less important than that in the counterflow flame, contributing only about 5% to fuel consumption in the premixed flames. In the non-premixed counterflow flame, the fuel mixture is pre-heated without oxygen as it approaches the flame surface, undergoing pyrolysis in which the direct decomposition of CH_3CHO is a vital pathway for fuel consumption. Compared to the counterflow flame experiments, the

premixed flame structure and the low pressure extend the lifetime of radicals and enhance the oxidation of various radicals and intermediates, increasing the production of the oxygenated intermediates such as methanol, ketene and acetone, as shown in Fig. 15(f)–(i).

Generally, both the counterflow flame and premixed flames can provide validation for the fuel consumption pathways under flame conditions. The experimental results of the non-premixed counterflow flame can provide more information about the part of hydrocarbon growth in the overall model, while the experimental results of the low-pressure premixed flames emphasize the aspects of the oxygenated intermediates.

4.3.4. Low-temperature oxidation in JSR

The present model, developed based on the previous model mainly by updating the high temperature reactions, has been validated against our previous experimental results of the low-temperature oxidation in a jet-stirred reactor [28]. Among the reactions related to the low-temperature oxidation pathways, only the H-abstractions on CH_3CHO by CH_3O_2 and CH_3CO_3 radicals have been modified. Therefore, the present model shows similar predictions to the previous one, as presented in Fig. 16. Only the comparisons for major species and five selected key intermediates which are important to the low-temperature oxidation are shown here.

The speciation results of acetaldehyde oxidation at higher temperature (950–1350 K) were measured by Dagaut et al. [8] in a jet-stirred reactor, which is also used to test the performance of the present model. Figure 17 is an example for the comparison between experimental and simulated results of $\text{CH}_3\text{CHO}/\text{O}_2/\text{N}_2$ oxidation at 1 atm with equivalence ratio of 1.61 and residence time of 0.08 s, which proves a good prediction ability of the present model.

5. Conclusions

In this work, the mole fraction profile of 31 species in a counterflow flame fueled by acetaldehyde at 600 Torr was measured by a photoionization molecular-beam mass spectrometer. Ignition delay times at 10 atm were measured in a rapid compression machine between 700–1100 K. A negative temperature coefficient behavior was observed, but its explanation is different from that for the NTC phenomenon in the JSR oxidation experiments, because the temperature range for the NTC behavior in the observed ignition delay times is too high for the production of the CH_3CO_3 radical, which is a key intermediate to explain the NTC phenomenon in the JSR oxidation.

An updated kinetic model is developed based on our previous kinetic model [28] to describe the oxidation of CH_3CHO . The current model has been comprehensively validated against experimental data obtained in this work, as well as the literature data [8,15,21,22,26,28], proving its good performance under wide ranges of temperatures (300–2300 K) and pressures (0.02–10 atm).

Based on the current model, the observed NTC behavior in the ignition delay times is explained by the different fate of CH_3 radicals at different temperatures. At low temperatures, the recombination of CH_3 radicals form stable C_2H_6 , which reduces the reactivity of CH_3CHO oxidation. As the temperature increases, C_2H_6 starts to be active and consumed, which restores the overall oxidation reactivity.

The present work provides a good start for further experimental and modeling investigations on the acetaldehyde combustion. More experiments are needed to explore the interesting NTC behavior in the ignition delay times under wider conditions to explore the influence of pressure and equivalence ratio, which will enrich our understanding about the low-temperature oxidation mechanism. For the model development, more efforts are needed on the O_2 -addition to the CH_3CO radicals, whose rate constant

still has large uncertainty. Theoretical work for more accurate kinetic parameters are also encouraged on the reactions related to the CH_3CO_3 radical, which is of kinetic significance in the low-temperature oxidation of acetaldehyde [28].

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2018.06.007.

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